SUNANDITHA B S

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Education

RNS Institute of Technology, Bengaluru CGPA: 8.9 / 10

B.E. Information Science & Engineering

Technical Skills

Languages: C, Python, Bash, SQL **Systems:** Linux, Docker, Embedded Systems, Git

Systems Engineering Experience

- Diagnosed two independent Linux boot failures via live-USB chroot recovery: a proprietary GPU driver corrupting the init sequence before the display manager could load, and an Intel VMD/RST storage controller loading out of sequence; resolved both through initramfs/GRUB rebuilds and kernel module reordering.
- Resolved a blocked ext4 partition expansion by working at the sector level, mapping GPT geometry around immovable NTFS $MFT mirror metadata that blocked every standard resize tool, safely reconstructing partition boundaries for a 60% capacity increase with zero data loss.
- Reverse-engineered a MediaTek Android device's boot trust chain (BROM → Preloader → LK → AVB) to flash a custom system image onto a locked bootloader, patching vbmeta verification flags and using fastbootd for logical-partition access.
- Root-caused a 4-layer networking failure chain (TLS, IPv6, DNS, socket permissions) blocking a containerized FastAPI deployment on Docker, isolating and resolving each layer independently before verifying the service end-to-end.

Projects

CANary: CAN Bus Intrusion Detection Framework July 2026 – Present

Lead Developer
Python • C • SocketCAN • Linux • ctypes • Rich • pytest

- Built an 8-module CAN bus intrusion detection framework in Python and C, simulating 3+ ECUs over a shared bus with priority-based arbitration and a native 15-bit CRC checksum bridged via ctypes.
- Designed 6 independent rule-based detectors covering ID spoofing, replay attacks, flooding, and malformed frames, producing 4-tier severity alerts and a live terminal telemetry dashboard.
- Validated the system against Linux SocketCAN on a virtual CAN interface, identifying and fixing a kernel-level frame loopback bug; covered by 47 passing unit and integration tests.

BICE: Behavioral Identity Continuity Engine Feb 2026 – Present

Lead Developer | Submitted to IEEE ICOSICS 2026 (Under Review)

- Built a per-device behavioral anomaly detection engine for IoT network telemetry, modeling DNS query entropy and port access frequency as streaming drift features without labeled attack data.
- Designed a constant-memory streaming pipeline with sliding-window Z-scores to score behavioral drift in real time across a 76-dimensional feature space.
- Authored a research paper detailing the detection methodology and evaluation against the N-BaIoT dataset, submitted to IEEE ICOSICS 2026 for peer review.

WatchTower: Adversarial Emulation & Telemetry Lab Dec 2025

Lead Developer

- Built a tiered adversarial emulation honeypot (public, employee, admin) using FastAPI to simulate attacker behavior across SQL injection, XSS, credential stuffing, and reconnaissance scenarios.
- Generated and analyzed 5,000+ telemetry events from simulated adversary activity, mapping behaviors to MITRE ATT&CK techniques to study detection opportunities across the attack lifecycle.
- Performed detection engineering on the collected telemetry, tuning alert logic to reduce false-positive volume and improve signal quality.

Additional

TryHackMe Global Top 5% | Winner, Escape and Exploit Cybersecurity Competition | Top 15, ZeroDay Arena CTF